

Research in Innovation, Digital Transformation, and Strategic Management

From Idea to Scientific Work (DBA-level, 5-hour interactive workshop)

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Agenda (5 hours total)

Module 1 (45')	What is DBA research? How it differs from consulting.
Module 2 (60')	From practice problem to research-relevant question.
Module 3 (60')	Building the theoretical foundation and identifying gaps.
Module 4 (75')	Choosing research designs that fit innovation/DT/strategy questions.
Module 5 (60')	Drafting a mini-proposal in teams and sharing.

How we'll work

- Highly interactive: short inputs, group work, report-outs.
- Focus on *clarity, feasibility, contribution*.
- Produce a 2-page mini-proposal by the end.
- Use the slide appendix for checklists and templates.

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Module 1 — What is research?

Research vs consulting deliverables

Learning goals: Understand the *scientific* part of applied research.

Consulting / managerial report

- Solves a client-specific problem.
- Emphasizes recommendations and ROI.
- Limited generalizability.

Scientific work

- Frames a question in existing theory.
- Produces evidence with transparent method.
- Contributes to *generalizable* knowledge.

Takeaway: DBA = bridge between practice problems and theory-driven explanations.

Case: From practice to research (AI in assessment)

Learning goals: See how a managerial issue becomes a research question.

Reference: <https://www.pnas.org/doi/epdf/10.1073/pnas.2425439122>

- **Practice problem:** Organizations disclose AI use in hiring/admissions; candidates know they are AI-assessed.
- **Research question:** Do people change their self-presentation when assessed by *AI vs humans*?
- **Core finding:** People present themselves as **more analytical** and **less intuitive** under AI assessment (*AI assessment effect*).

Takeaway: Research turns a live managerial concern into a *testable* contribution.

Why this is research

Scientific angle

- Theory-driven (impression management; lay beliefs about AI).
- Pre-registered, multi-study evidence (lab, field, behavioral tasks; $N=13,342$).
- Identifies a mechanism (analytical-priority lay belief) and boundary patterns.

Managerial angle

- Hiring validity & fairness implications.
- Disclosure/transparency policy interacts with behavior (EU AI Act context).

Takeaway: Both scientific rigor *and* practical relevance.

Activity: Vote & justify.

1. Market sizing for a new digital service.
2. Interview-based study on why firms resist cloud migration.
3. A/B test of a feature in an innovation platform.

Takeaway: Not the *task* but the *framing*, *method*, and *contribution* make it research.

Module 2 — From problem to research question

From practice problem to problem statement

Learning goals: Rewrite managerial pain points as researchable problems.

- Practice problem (raw): “Employees don’t adopt our AI tools.”
- Problem statement (refined): “Managers perceive AI systems as opaque, undermining trust and adoption.”
- Scope: context (industry), actors, time, outcome.

What makes a good research question?

- Specific and *answerable* (who/what/when/where).
- Grounded in *literature* and *gaps*.
- Feasible data/method within DBA constraints.
- Speaks to *innovation/DT/strategy* debates (theory contribution).

Takeaway: Aim for: novelty, clarity, feasibility, contribution.

From topic to research question (worked example)

Following our worked example (see referenced paper *here*)

- **Topic:** AI in HR/Admissions.
- **Problem statement:** When told an *AI* is the assessor, candidates might strategically adapt their self-presentation.
- **RQ:** How does disclosure of *AI* (vs human) assessment change applicants' *analytical vs intuitive* self-presentation?
- **Mechanism (candidate):** Lay belief that AI prioritizes analytical traits $\Rightarrow$ behavior shifts.

Example hypotheses from the case

- **H1 (main effect):** Disclosure of *AI* assessment increases *analytical* self-presentation and decreases *intuitive* self-presentation (vs human).
- **H2 (mediation):** The effect is mediated by the *analytical-priority lay belief* (belief that AI values analytical traits).
- **H3 (attenuation):** Correcting/challenging lay beliefs (“consider-the-opposite” prompts) reduces or reverses H1.

Takeaway: Clear RQ → directional hypotheses → testable mechanism.

Activity: In teams, pick a topic and reframe it into a research question.

Some ideas (not necessary to follow these)

- AI augmentation and trust
- Digital transformation resistance
- Platform/ecosystem governance

Deliverable: 1 problem statement + 1 research question per team.

Module 3 — Building the theoretical foundation

Why theory?

- Explains *why* and *how*, not just *what*.
- Connects your case to wider knowledge.
- Enables testable hypotheses or propositions.

Common lenses in this track

Innovation / DT

- Dynamic capabilities
- Technology acceptance / user trust
- Socio-technical systems

Strategy / Ecosystems

- Platform governance / orchestration
- Institutional theory
- Resource-based / complementarities

Mapping a theory to a testable model

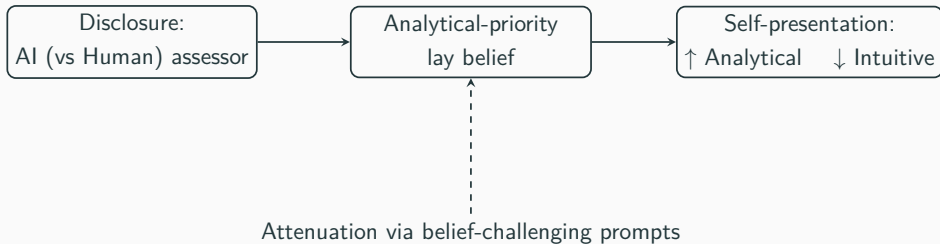
Activity: Sketch variables and arrows.

- Example: AI opacity $\rightarrow$ trust $\rightarrow$ adoption (mediation).
- Consider moderators: experience, data criticality, industry.
- Convert to hypotheses: H1, H2, H3.

Theoretical lenses behind the case

- **Impression management / self-presentation:** People strategically adapt to assessor standards in high-stakes assessments.
- **Lay beliefs about AI:** Common belief that AI is analytical/rational and less emotional/intuitive; this guides behavior.
- **Dual-process backdrop:** Analytical vs intuitive styles as a fundamental dimension.

Mechanism map (simple)



Evidence: randomized experiments & behavioral ranking tasks support mediation.

Module 4 — Choosing a research design

Design follows question

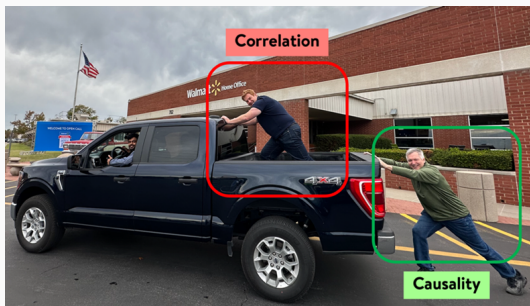
- **Exploratory:** Multiple-case, interviews, grounded theory.
- **Confirmatory:** Surveys, field/natural experiments, archival.
- **Mixed methods:** Triangulate for credibility.

Takeaway: Method *serves* the question; avoid method-driven projects.

Correlation vs. Causation

Correlation: A statistical relationship between two variables where changes in one variable are associated with changes in another.

Causality: refers to a direct cause-and-effect relationship where changes in one variable lead to changes in another.



Correlation does not imply causation!

Laboratory Experiments

Abstract task, Student populations, Imposed rules



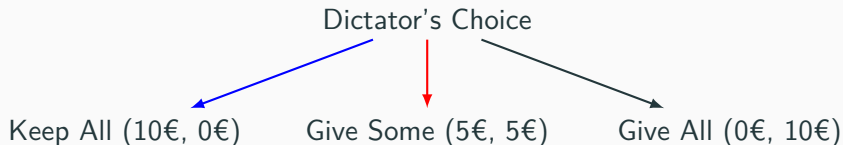
- From 15 to 30 terminals + facility with separators
- Most used software: Ztree (.exe) and oTree (python + HTML)
- University Recruitment platform (e.g., ORSEE)

Some Interesting Examples of Lab Exp.

Exp. Econ is based on *game paradigms* : strategic situations in which we can use game theory to predict behavior.

Example: to measure individuals' generosity

- One could ask "How often do you help others?" (World Value Survey style)
- An experimentalist would say: **Dictator Game**



Prosociality & Environment Attanasi et al. 2025

Same paradigm, but different story behind: increase **voluntarily donations** to environmental organizations using AR visualizations



2 Conditions: Control (only Text), Treatment (Text + AR)

Natural Field Experiments



Figure 1: Reducing energy consumption using smart meters (Allcott 2011)

Natural environment, and subjects are not informed they are part of an experiment

Objective: Reducing carbon footprint of internet data use

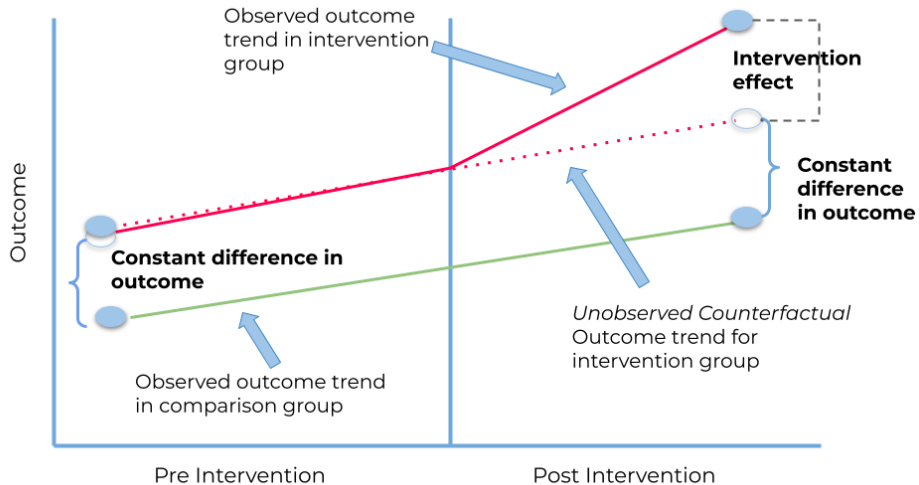
- 4/5G produces 4x more CO₂ than Wi-Fi

Experiment in a nutshell:

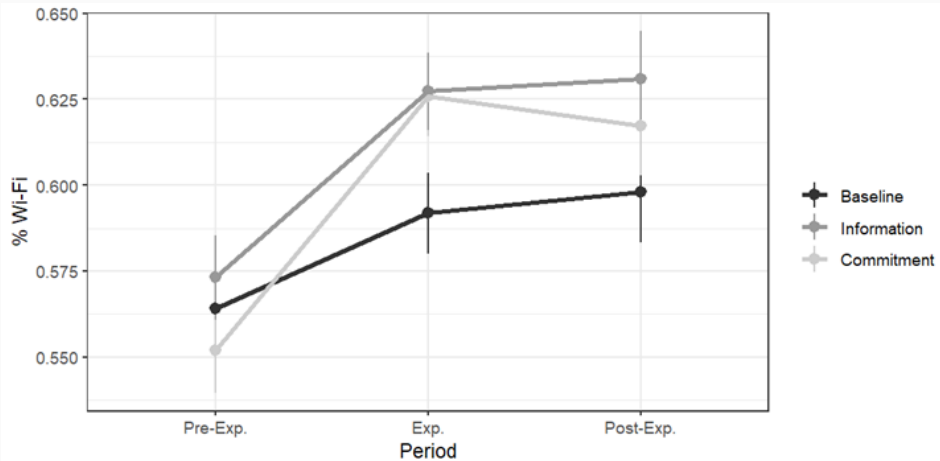
- Randomization across 3 conditions (between-subject)
- Each condition differ in how we "push" participants to reduce mobile consumption
 - Control (just observe)
 - Information (provide info on consumption consequences)
 - Commitment (set consumption goals)
- Use of a smartphone app to monitor data consumption
 - App retrieves data consumption **before** installation
 - so we have **Pre-** and **Post-**Intervention

Why I chose this example

Difference-in-Difference Design



Results



Other examples

Productivity - **Gossnell et al. 2021:** Testing managerial practices on Airlines Pilots' Behavior and Productivity

Electricity consumption - **Allcott 2011:** Testing whether social norms influence electricity consumption

Our worked example - Design overview (multi-method, pre-registered)

- **Pilot field (Upwork):** Freelancers disclose being more analytical when told AI assesses them. Small effect, real setting.
- **Study 1:** Randomized AI vs human; *more analytical, less intuitive* approaches under AI; mediation via lay belief.
- **Study 2:** Representative US sample; robustness; effect stronger among younger cohorts (floodlight JN $\approx$ 62y).
- **Study 3:** 2-stage design; switching assessors changes self-presentation; potential selection consequences by thresholding.
- **Study 4:** *Behavioral* ranking (attributes): AI $\Rightarrow$ higher rank of analytical traits (lower rank sums).
- **Studies 5a/5b:** Challenging the belief attenuates or reverses the effect.
- **Study 6:** Joint AI+human disclosure reduces (not removes) the effect.

Takeaway: Designs match the RQ & mechanism; triangulation builds credibility.

Results snapshots to discuss

- **Main pattern (Study 1):** AI $\Rightarrow$ $\uparrow$ analytical ($d \approx .44$) and $\downarrow$ intuitive task approach; mediation by belief.
- **Behavior (Study 4):** In a consequential ranking task, participants prioritize analytical attributes under AI (Wilcoxon $z = -4.46$, $p < .001$).
- **Policy-relevant (Study 3):** With a selection threshold (e.g., ≥ 6 on 7-pt scale), $\sim 27\%$ would be selected under AI but not under human assessment.
- **Mitigation (5a/5b):** “Consider-the-opposite”/“AI can read emotions” prompts reduce or reverse the shift.
- **Hybrid (Study 6):** Saying “AI screens, human decides” partially attenuates the effect.

Data planning checklist

- Units of analysis, sampling frame, access feasibility.
- Measurement (constructs, items, proxies), pretesting.
- Validity, reliability, bias mitigation (CMV, selection, omitted vars).
- Ethics: consent, privacy, reproducibility of code/data.

Activity: Match questions to designs.

Ways to fail in the field

Main reasons leading to failing in the field

- Inappropriate research design
- Partner organization challenges
- Technical design flaws
- Low participation rate

From book by Karlan & Appel *"Failing in the Field: What We Can Learn When Field Research Goes Wrong"* + Personal Experience

1. Inappropriate research design

- Context matters: Research should be conducted in settings where the problem exists and the intervention is relevant.
- Timing issues: external events like politics or weather can disrupt research
- Feasibility: Ensure infrastructure and resources are available
- Knowing when to stop: do not be afraid to drop the study

Tip: Enlarge your toolbox

- **Qualitative methods:** Use interviews, focus groups, or field observations to understand the context and refine your research question.
 - **Combine and Iterate:** Combine methods and let early qualitative findings inform your quantitative design and vice-versa.
- **Descriptive Evidence:** No need for causality all the time! Stylized facts are precursor of bigger ideas

An example: Guido et al. 2024

Problem. Switching from mobile to Wi-Fi data consumption (offloading)

Before going to the field and design interventions, get familiar with the problem at hand!

Interviews with Providers and Users + Preliminary Survey: knowing whether people know about offloading and what interventions would be most effective

- more than 50% of subjects did not know about the offloading problem → Inform them!
- Mobile consumption correlates with impatience → Improve self control (commitments)

Co-design sessions: group-up approach to designing experiments

Did you know that mobile networks produce up to 4 times more CO₂ than Wi-Fi networks?

According to your declaration of **Mobile** data consumption, you will produce:

2290g of CO₂/per month

If you had used **Wi-Fi** only with the same amount of consumed data, you will produce only:

536g of CO₂/per month

By **switching** to **Wi-Fi** network, your **CO₂ emissions will be reduced** by a quantity equivalent to:

32h electricity heating	
219 Km in High speed train	
2928h of LED bulb on	

As a reminder, here is your breakdown of last week's data consumption:
38% via Wi-Fi

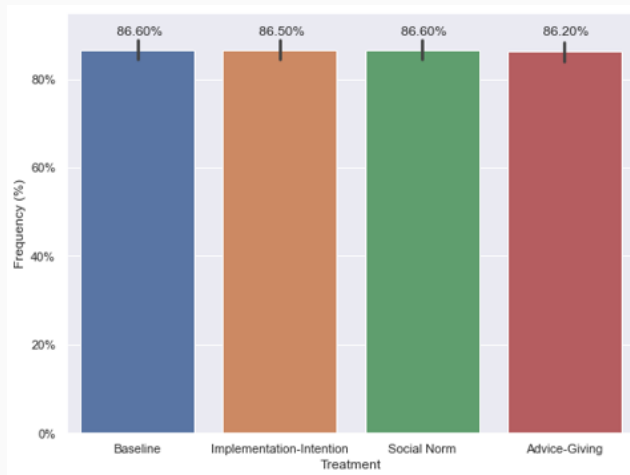
Example 2: Romaniuc, Guido et al. 2024

Problem. Nudging voter participation among young adults

Co-design sessions with experts



Yet, no pre-experimental survey on baseline participation rates



Tip: Enlarge your toolbox

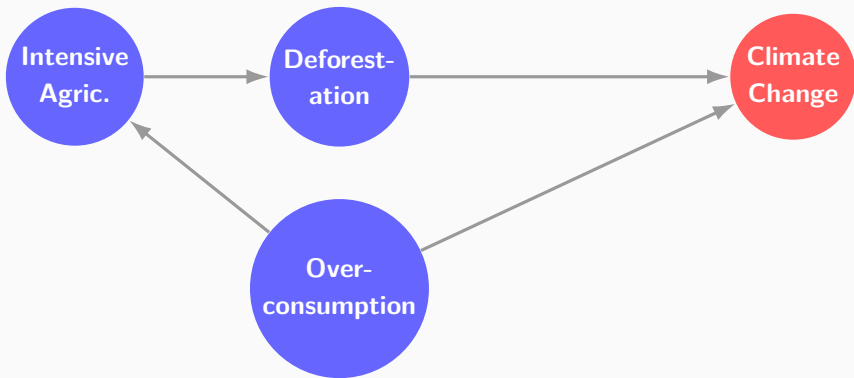
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An Example: Narratives of Climate Change

Ongoing work with Ph.D.s visiting PSB (Bea & Dany)

- **Factors causing climate change:** We use open-ended questions to elicit narratives on climate change
- Survey 10 countries (Europe, US, Asia) 3000+ respondents
- Transform narratives into graphs

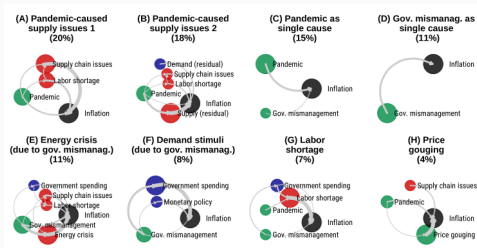
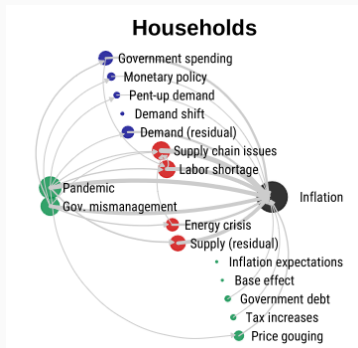
Causal map example: narrative structure



“Overconsumption drives climate change, partly via higher agricultural demand leading to deforestation.”

An Example: Narratives of Climate Change

Find patterns, stylized facts and competing narratives



2. Partner organization

- It is hard to collaborate with organizations to implement programs and gather relevant data.
- Organizations are often afraid of experimenting (never say the “E” word)

My view:

- Show the advantages of using mixed methods
- Use experiments as a complementary tool to measure opportunity costs

3. Technical Design Flaws

- **Survey Design Errors:** Question wording, response scales and question order
- **Survey Length and Relevance:** Tendency to add unnecessary questions
- **Statistical Power and Sample Size**
- **Randomization Protocols**

4. Low participation rates

- Distinguish between *during intake* and *after treatment assignment*
- *During intake*: less participants than expected
 - sampling, hard-to-reach participants
 - wrong incentives
 - trust in the researchers or organizations involved
- *After treatment assignment*: potential bias in estimates
 - treatment makes harder to stay in the study?

Example: Loan Repayment in Nigeria (Guido & Romaniuc, in progress)

Trust towards researchers is important!

Objective. Measuring external validity of laboratory measures in the context of loan repayment

- Bank kindly provides customers' contacts
- Researchers (w/ local RA) send invitation emails
- Response rate: 5% !!!
- They thought we were scamming...
- Solutions: Involvement of local university + stricter protocol (phone calls with pre-defined script)

Future directions & Hot topics

3 Waves

- Wave 1 (90'): Testing (classic) theories

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 - Oh, we are not rational!

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- Wave 4: ?

Hot topics

Climate change, Green behavior and sustainability

- Reduce or change polluting behavior
- Correct misperceptions and wrong beliefs

Cognitive foundations in decision making

- The ways in which people attend to, remember, aggregate and trade off information
- Mental models and Narratives
- Motivated Reasoning

Digital Technologies

- Role of AI in decision making
- Digital Addiction (young adults)
- Environmental consequences of extensive use

Sum Up!

- Beh. Sciences and Experiments: a fun way to get answers
- Mixed methods are the future and we can learn from each other
- I hope this talk will stimulate new research ideas and initiatives

Module 5 — From idea to mini-proposal

Team workshop (deliverable)

Activity: Produce a 2-page mini-proposal per team.

1. Title + 5–8 keywords.
2. Motivation & problem statement.
3. Research question(s) / hypotheses or propositions.
4. Theoretical background (2–3 key refs).
5. Proposed design: method, data, measures, analysis.

Mini-proposal template (2 pages)

Title	Concise, informative; includes context and main mechanism.
Keywords	5–8 terms (innovation, DT, strategy, platform, adoption, trust, governance, etc.).
Motivation	Why this matters for managers/society; succinct context.
Problem	Precise statement of the issue; define scope and actors.
RQ/Hypotheses	1–2 focused RQs; if confirmatory, 2–4 hypotheses.
Theory	2–3 anchor papers/lenses; simple variable map.
Design	Method (case/survey/experiment/archival), sampling, measures, analysis.
Ethics	Consent, privacy, reproducibility plan (code/data).
Feasibility	Access to sites/data, timeline, risks/mitigations.
Contribution	To theory (gap addressed) and to practice (actionable insight).

Mini-proposal (filled example from the case)

1. **Title:** AI Assessment and the Analytical Self.
2. **Motivation:** Transparency laws (e.g., EU AI Act) make AI assessment salient; candidates may strategically adapt.
3. **RQ:** How does disclosing AI (vs human) assessment change self-presentation on the analytical–intuitive dimension?
4. **Theory:** Impression management + lay beliefs about AI's analytical priority.
5. **Design:** Randomized disclosure; measure/observe task approach or attribute rankings; test mediation; try lay-belief debiasing prompts.
6. **Contribution:** Identifies a systematic behavioral bias induced by AI disclosure; offers mitigation levers (messaging, hybrid processes).

Another example (more comprehensive)

Pitcher Grid by Faff

Pitch and peer feedback

- 2-minute pitch per team.
- 2-minute peer feedback: clarity, feasibility, contribution.
- Instructor sums up patterns and suggestions.

Wrap-up

Path to a complete DBA thesis

1. Idea $\Rightarrow$ problem statement $\Rightarrow$ research question.
2. Theory map $\Rightarrow$ hypotheses/propositions.
3. Design $\Rightarrow$ data $\Rightarrow$ analysis and testing.
4. Interpret $\Rightarrow$ contribution to knowledge & practice.

Takeaway: Keep iterating: refine question, theory, and feasibility in lockstep.

What to do next (after today)

- Expand mini-proposal into an A01-style document.
- Build a minimal reading list (10–15 core papers) and summary table.
- Validate access to data and pilot your instrument/protocol.
- Draft a Gantt with milestones to analysis and write-up.

Appendix B: Research question quality rubric

Dimension	Indicators
Clarity	Actors, outcome, mechanism, context stated.
Novelty	Extends, refines, or challenges extant findings.
Feasibility	Data access realistic within time/budget; measurable constructs.
Theoretical fit	Linked to a recognized lens; enables testable claims.
Practical relevance	Addresses real managerial decisions/trade-offs.

Appendix C: Example hypothesis framing

- H1: Greater *perceived AI opacity* is associated with lower *managerial trust*.
- H2: Managerial trust *mediates* the effect of opacity on *adoption intentions*.
- H3: The negative effect of opacity on trust is weaker when *explanations are provided* (moderation).

Appendix D: Data & method quick guide

Design	When to use / Notes
Multiple-case study	Explore processes, mechanisms, context; rich evidence; theoretical sampling.
Survey (cross-sectional)	Test associations; ensure measurement validity; handle CMV bias.
Field experiment	Estimate causal effects; pre-registration and ethics essential.
Archival data	Objective indicators; careful identification strategy.
Mixed methods	Complement weaknesses; sequence for explanation and testing.

Appendix E: Further reading (starter list)

XXX

- Innovation/DT: *Technological Forecasting & Social Change*; *MISQ*; *Information Systems Research*.
- Behavioral sciences (at large): *Nature Comms*, *PNAS*, *Journal of Economic Behavior and Organization*
- Methods: Yin (case studies), Creswell (mixed methods), Field (regression), Trochim (research design).

Questions? Ready to draft your mini-proposal!